

that ensure a stable and efficient end product.

A major demand driver of flexible electronics is the consumer electronics sector. This predominantly includes OLED-based display devices. Flexible OLED lighting solutions are expected to be increasingly adopted for automotive applications. These are easy to integrate, provide improved luminosity and offer optimisation in saving space and cost. Growing interest in flexible OLED lighting from major automobile manufacturers is also expected to boost this high rate of adoption.

Flexible batteries are also expected to be rapidly adopted for consumer, healthcare and industrial applications. Improvements such as rechargeability will aid longer operational times in various applications.

Consumer electronics is set to retain this domination with the inclusion of other innovative products.

Other areas that will see a growth in the adoption of printable electronics are printed batteries, healthcare and transport.

Flexible electronics has made possible a whole range of products that were hard to fathom with traditional electronics. For example, printed solar panels, printed memories, flexible and printed displays and disposable printed batteries are some of the products being realised using flexible and printable electronics.

Further, the light weight and form factor of the products have allowed flexible electronics to penetrate into other innovative applications like smart textiles.

Flexible displays account for a major share of the flexible electronics market for consumer electronics. Flexible displays are prominently being adopted for smartphones, smart wearables and smart television applications. Leading smartphone brands such as Apple and Samsung are playing a vital role in the growth of the flexible electronics market for flexible displays.

Continuous innovation and development of advanced organic electronic technologies is a key factor that is likely to augment the growth of the global organic electronics market in

the near future. Factors such as rising disposable incomes of the population and increased spending on innovative technologies are fuelling the demand for organic electronics across the globe.

In addition to this, new applications of organic materials in electronics is likely to intensify the size of global organic electronics market in the future. Additionally, implementation of organic materials in display technologies such as smartphones, tablets, televisions and laptops are predicated to enhance the growth of global organic electronics market in upcoming years.

In terms of geographic regions, Asia-Pacific (APAC) dominated the flexible electronics market, accounting for the largest market share in 2017, according to a Market and Market report. The region is likely to be the fastest-growing market for flexible electronics during the forecast period. APAC is the hub for manufacturers and customers of flexible OLED panels, with Samsung and LG being the leading companies that deploy flexible displays in smartphones and televisions.

In APAC region, flexible sensors are increasingly being used for healthcare applications. The region has witnessed an increase in the installation of thin-film photovoltaics in recent years. Their deployment is expected to increase steadily in China. With Japan and India being prominent countries in the adoption of thin-film photovoltaics, other countries in the region are also shifting towards renewable energy resources.

India advantage

Flexible and printed electronics, although a young technology, offers revolutionary flexibility in both literal and figurative senses of the word. Using flexible circuitry, manufacturers have the potential to create devices with functionalities that conventional electronic devices cannot achieve. These include displays, sensors and printed memories that can be bent, wrapped, rolled and stretched to give manufacturers the potential to make great strides in reducing cost, weight and power consumption of electronic

devices. Progress will have to be made in reducing and optimising development cycles, as applications for the technology will continue to rapidly emerge in the coming years.

India is in an advantageous position to explore these segments because of the availability of multi-disciplinary experts and skilled workforce in the areas of electronics, physics and material science. Moreover, reputed academic and research institutes are already engaged in high-quality basic research in a variety of printed and potentially printed electronic and electric devices.

For example, Defence Laboratory, Jodhpur and Indian Institute of Technology, Delhi (IIT-D) are working on photovoltaics. Indian Institute of Technology, Kanpur (IIT-K) has set up National Centre for Flexible Electronics (NCFlexE) with an outlay of ₹ 1.33 billion. Out of the total outlay, ₹ 1.11 billion will be funded by Ministry of Electronics and IT (MeitY) and the contribution of IIT-K will be ₹ 218.7 million. The objective of NCFlexE is to spur the development of an ecosystem for flexible electronics in the country.

Development of flexible electronics in India is in line with Digital India vision of the government. This serves to develop the overall Electronics System Design and Manufacturing (ESDM) sector to achieve net-zero imports by 2020, thereby, facilitating India's image as an exports hub in the ESDM sector. It is with this objective that an electronic development fund is set up as a fund of funds to participate in professionally-managed daughter funds. This will, in turn, provide the risk capital to companies developing new technologies in the areas of electronics, nanoelectronics and information technology.

However, the Indian electronics industry is yet to make use of this opportunity to get a foothold in the emerging market for large-area flexible electronics. Considering that there are only a few well-established players in flexible electronics sector currently, and even those who are present have been in it for a relatively short duration, the Indian industry must explore this segment. **EFY**